Kushal Agrawal

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Education

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Computer Engineering — *Aug 2026 – Dec 2027*

University of Wisconsin–Madison — Madison, WI
Bachelor of Science in Computer Engineering — *May 2026*

Coursework: Operating Systems, Data Structures & Algorithms, Distributed Systems, Computer Architecture, Object Oriented Programming, High Performance Computing, Microprocessor Systems, Probability, Digital System Design

Experience

Google (Capstone) — Madison, WI
Software Engineer Intern — *Feb 2026 – May 2026*
- Built IoTriage, a full-stack tool that scans a local network for IoT devices, identifies known vulnerabilities via the NIST NVD, and generates Gemini remediation guidance (Python, Nmap, Express, React).
- Built a rate-limit-aware NVD client with severity and false-positive filtering for accurate vulnerability rankings.

University of Wisconsin–Madison — Madison, WI
Undergraduate Teaching Assistant — *Sep 2023 – May 2026*
- Led office hours for 850+ students in Digital System Fundamentals, Machine Organization and Intro to Comp. Eng.
- Instructed students in CS 352, CS 354, and CS 252 topics including digital logic design, processor architecture, RTL design, finite state machines, timing analysis, pipelining, C/assembly programming, and memory hierarchy.

Endress+Hauser Group — Greenwood, IN
Embedded Software Engineer Intern — *May 2025 – Dec 2025*
- Engineered a Dockerized full-stack Netilion IIoT simulator (React/TypeScript, Python/FastAPI) with 30+ REST endpoints, 5 simulation patterns and 200+ health codes, streaming real-time data for hardware-free demos.
- Characterized Bluetooth 6.2 performance on the nRF54L15 via Channel Sounding and multi-device tests, analyzing ranging accuracy, packet reliability, and RF congestion effects for the SGC200 Bluetooth-to-cloud gateway.

Software Engineer Intern — *May 2024 – Aug 2024*
- Built algorithms to analyze 1,100+ Salesforce dashboards for duplicates, then deployed a TypeScript/React naming web tool—integrated it as an extension—and proposed it across 10 departments, potentially saving $40K annually.
- Designed a custom analog front-end and microcontroller interface to generate and read a 4–20 mA signal from an industrial temperature probe, enabling accurate current-to-temperature conversion for real-time IoT visualization.

Projects and Leadership

FileHawk.net–Local Semantic Search | *Electron, Python, ChromaDB, SentenceTransformers* | filehawk.net
- Designed and shipped an on-device semantic file retrieval app (Electron/React + Flask) that returns relevant local results in <1s, with all embedding inference running locally—no cloud path for core search.
- Engineered a retrieve-then-rank search pipeline: SentenceTransformers (MS MARCO/All-MiniLM) embeddings queried via ChromaDB, then hybrid ranking signals to score semantic matches across 25+ file types.
- Built incremental indexing (SHA-256 change detection + filesystem watching) that reindexes only changed files.

High-Frequency Currency Arbitrage Engine | *C++, CUDA, OpenMP, Python, CMake* | GitHub
- Built a parallel FX arbitrage engine analyzing 86,400 per-second timestamps across 19 currency pairs, modeling markets as dynamic graphs for real-time detection.
- Detected 680 arbitrage opportunities, achieving 0.71% simulated daily ROI with portfolio tracking.
- Leveraged parallelization to cut detection latency by 3× and achieve a 3.5× speedup in arbitrage cycle processing.

FUSE-Based Linux Filesystem | *C, FUSE, RAID 0/1, Python, Bash, GDB* | GitHub
- Built a user-space filesystem in C using FUSE, implementing inode-based storage, indirect block mapping, bitmap allocation, superblock-based metadata, hierarchical directories, RAID 0/1, and automated end-to-end tests.

Executive Board Member | *Engineering Expo* — Sep 2022 – May 2026
- Organized a large-scale STEM outreach event at UW-Madison, attracting 5,000+ visitors annually.

Technical Skills

Languages: C, C++, Python, Java, C#, TypeScript, JavaScript, Bash, SQL, x86 Assembly
Web & Application: React, Node.js, Electron, Next.js, FastAPI, Flask, REST APIs, HTML/CSS, Tailwind CSS
Machine Learning: PyTorch, SentenceTransformers, Hugging Face, ChromaDB, NumPy, Pandas, Matplotlib
Systems & Tools: Linux, Git, Docker, CMake, CUDA, OpenMP, FUSE, GDB, Slurm, AWS, GCP, Postman